Final Report

Claire Standley

2026-04-09

Motivation:

The central question motivating this project is whether the adoption of agritourism liability laws causes farmers to expand direct to consumer (DTC) activity. However answering this question causally would need the timing of law adoption to be unrelated to pre-existing trends in DTC agriculture aka the parallel trends assumption. In this setting that is very unlikely to hold. States with growing agritourism sectors may be more likely to adopt liability protections precisely because their farmers are already engaging in these activities and pushing for legal certainties to be in place. Alternatively those states with larger agritourism communities already may have systems in place where they no longer see the demand from farmers to adopt laws. If either of these is the case comparing counties in adopting states to counties in non adopting states would conflate the effect of the laws as there are likely underlying systematic differences between the state and the agritourism industry in those states biasing estimates.

Due to this I have done my econometric analysis as a descriptive regression that serves as a first step toward causal inference rather than a causal claim in its own right. The regressions quantify the conditional association between law adoption and local food market activity after controlling for county and year trends. The goal is to describe what patterns exist in the data, state their magnitude and fit, and use this as a foundation for a more credible identification strategy. Basically this is serving as a first step in a larger question about small farm survival rates. The analysis also looks at the sensitivity of results to specific features of the control group that may bias estimates in a particular direction. As the research question is fundamentally causal the descriptive findings are interpreted through that lens but focus on limitations of the current design and why although using language similar to causal are NOT causal.

One factor that may influence the growth of local food markets is the legal environment surrounding agritourism. Agritourism includes activities such as farm tours, pumpkin patches, corn mazes, and on farm events that may attract visitors to agricultural operations. Other papers have observed clustering effects regarding agritourism and direct to consumer sales in the United States and found these two methods of income to be closely related in most cases. However, farms that host visitors may face liability risks if guests are injured while visiting the property. Starting with Kentucky in 2002, many states adopted agritourism liability laws which limit the legal liability of farmers hosting agritourism activities under certain conditions. By reducing the risk of lawsuits, these laws may encourage farmers to expand agritourism activities and attract more visitors. Increased visitor traffic also increases opportunities for farms to sell products directly to consumers.

Why This is a Causal Question:

The research question, does agritourism liability law adoption increase DTC agricultural activity, is a causal question rather than only descriptive or predictive. A descriptive question would be about what the association between these variables looks like without asking whether one drives the other. Predictive would be whether law adoption status can be used to forecast future DTC activity. This is well outside the scope of my data and likely not a question that would be viable in any setting. The question here is looking at the effect of a policy intervention. Would a state that did not adopt a law have seen different DTC outcomes if it had adopted one is the core of the analysis. This counterfactual framework is what makes this a causal question. The regression analysis to follow cannot fully answer the question however and has many many limitations but it provides the descriptive foundation for a potential causal analysis in the future.

Methods:

Econometric analysis:

To quantify the relationships observed in the exploratory analysis, I estimate linear regression models of the form:

$$log(1 + Y_{ist}) = \alpha_i + \gamma_t + \beta * Treated_{st} + \epsilon_{ist}$$

where i indexes counties, s indexes states, and t indexes years. The outcome Y is either new farmers market registrations per county per year or total farms reporting value added sales per county per census year. Both outcomes are log transformed using log(1 + Y) to address right skewed distributions and accommodate counties with zero counts consistent with the transformation applied in the exploratory analysis.

Treated st is a binary indicator equal to 1 if state s has adopted an agritourism liability law by year t, and 0 otherwise. Adoption years were compiled from the National Agricultural Law Center's publicly available records of state agritourism liability statutes beginning with Kentucky's adoption in 2002.

alpha_i represents county fixed effects which absorb all time invariant differences across counties such as differences in population density baseline agricultural character and regional food culture. gamma_t represents year fixed effects which absorb shocks common to all counties in a given year such as nationwide growth in interest in local food or potential changes in costs to USDA registration systems. Standard errors are clustered at the state level in all in all county level models because the treatment variable varies at the state rather than the county level .

The coefficient of interest (beta) describes the average within county difference in the outcome associated with operating under an agritourism liability law after controlling for county characteristics and year trends. It is interpreted as conditional association not a causal effect. To assess the robustness of the main results a set of specifications progressively excludes high production states without agritourism liability laws (Connecticut, California, and Massachusetts) which rank among the top five states by mean value added farms and market operations per county but contribute to the control group. Their potential structural or market advantage in DTC agricultural may inflate the control group mean and compress beta downward. Essentially that this group may systematically differ from the treated group and not serve as a good control since they are so much more productive than most. A state year panel is also estimated as an alternative aggregation that matches the level at which treatment is assigned.

Data Structure and Decision Not to Merge:

Two outcome variables are analyzed in separate regressions and are intentionally not combined into a single data set. The USDA Local Food Directory provides annual data from 2019 to present on new farmers market registrations per county, but it records only new registrations in a given year rather than total active markets. Since markets do not re-register annually this variable systematically understates true market prevalence and is more accurately understood as a measure of directory entry rather then market activity. The USDA Census of Agriculture provides data every five years from 1997 to 2022 on the total number of farms reporting value added sales per county a more comprehensive measure but lower frequency of reporting.

By combining these data sets by county and year the data set would be restricted to the single overlapping year 2022 collapsing approximately 18000 observations to a little over 2000 and eliminating virtually all panel variation. If the data sets were just joined on county then the understating of the Food Directory data would be a mismatch for the Census data reported because the measures of reporting number of operations are systematically measuring different things. Keeping them separate preserves the value of each and allows them to speak of different dimensions of DTC agricultural activity across different time horizons. Results from the two models should be read as complementary rather than redundant.

Data Processing:

I started by loading in my 3 data sets (USDA census data, ag law data, and farmers market data). All data sets were cleaned per the data cleaning checklist from the course. They were cleaned for consistency in variable naming and formatting first. I then standardized capitalization, and converted years to numeric. Duplicate rows and empty observations were removed to avoid double counts. I merged the data sets using county and year as the primary keys for county-level data, while the state laws was expanded into a panel for each state-year combination. This allows me to capture variation in farmers market prevalence and farm participation over time while linking outcomes to the timing of the state agritourism laws. The unit of analysis for the merged data sets is the county-year for the farmers market and census data, with state-level law characteristics attached to each county-year observation.

Transformations, Missing, and Extreme Values:

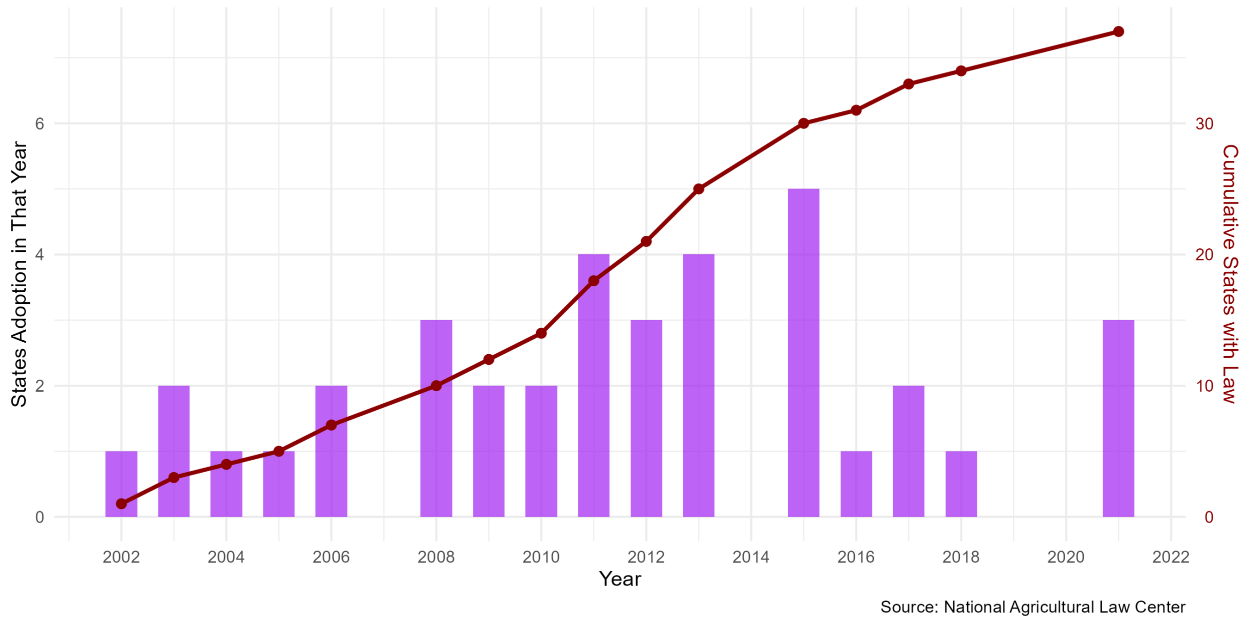
Several transformations were applied before moving into visualization. The number of farmers markets and total farms both exhibit right skewed distributions with a small number of counties having very large values relative to the majority. Due to the skewness if a normal linear regression in levels is used it can be heavily influenced by extreme observations leading to unstable estimates and heteroskedastic residuals. To address this concern I applied a log transformation using log(1 + x) which compressed the upper tail of the distribution but preserved observations with zero values. This reduced the influence outliers had and helps stabilize the variance of the error term making OLS assumptions more likely to hold. Also the log transformation will allow more meaningful intepretations for coefficients as they will be in percentage changes. Extreme values therefore are not overly influential in my data but don't get entirely dropped either. For categorical grouping, the number of statutory codes in state law was binned into meaningful groups at natural breaks to see how the law complexity could impact direct-market operations. Missing data was address in cases of the census data such as (D) which were converted to NA values as those values were not listed therefore not applicable as a value. The D is suppressed data withheld for individual reasons therefore could not be used. For the variable capturing farms with value-added sales, remaining missing values were replaced with 0 where appropriate (no instances that I could see but coded in case of values). I checked extreme values at the 50th, 90th, 95th, and 99th percentile to ensure no removal was necessary. Since our log transformation had already been done and no 99th percentile values were substantially large I did not remove values for this reason. Hopefully every decision I made was convey well in my reasoning behind the decision and that the data set cleaned reflects true participation in value added agriculture/programs.

Results:

Visualizations:

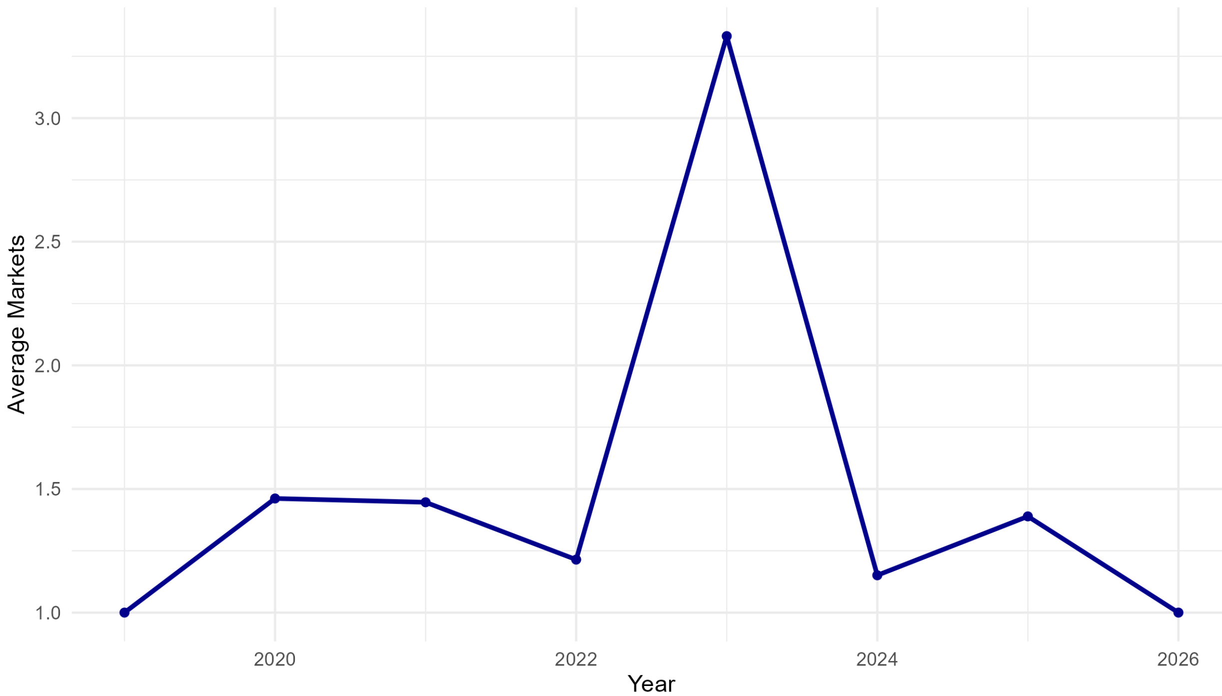
Figure A1. Agritourism Liability Law Adoption Over Time

Bars = new adoptions per year | Red line = cumulative states with law



This figure is important for understanding our independent variable of the limited liability agritourism laws. It shows the time trend with the stagger adoption by states. The darker line corresponding to the axis on the right shows the cumulative states with laws and how that grew over time. The bars down below show the number of states that adopted laws in that year. Clearly there was kind of a spike in larger adoption trends from 2011-2015 with 2015 being the largest year of adoption. It is quite interesting that we don't see any adoption in 2014 however given the large years around it and makes me wonder if something else was happening around that time.

Figure 1: Average New Registration of Farmers Markets per County Over Time



Shows overall trend in farmers market prevalence.

This figure illustrates the trend in average farmers markets in counties over time. It shows a clear increase in farmers markets in 2023 on average. This could be due to a push towards registration with the given site as individuals are the ones that provide the data. After registering in 2023 individuals did not have to register again so likely we saw a push for registration in 2023 and then the apparent fall is due to those farmers markets still being open but not necessarily re-registering. The overall outcome of interest is number of farmers markets for the farmer's market data or farms that participate in value added for the census data. This graph shows the overall trends of this outcome variable in the farmer's market data set over time.

Figure 2: Average Number of Farms with Value-Added Sales Over Time

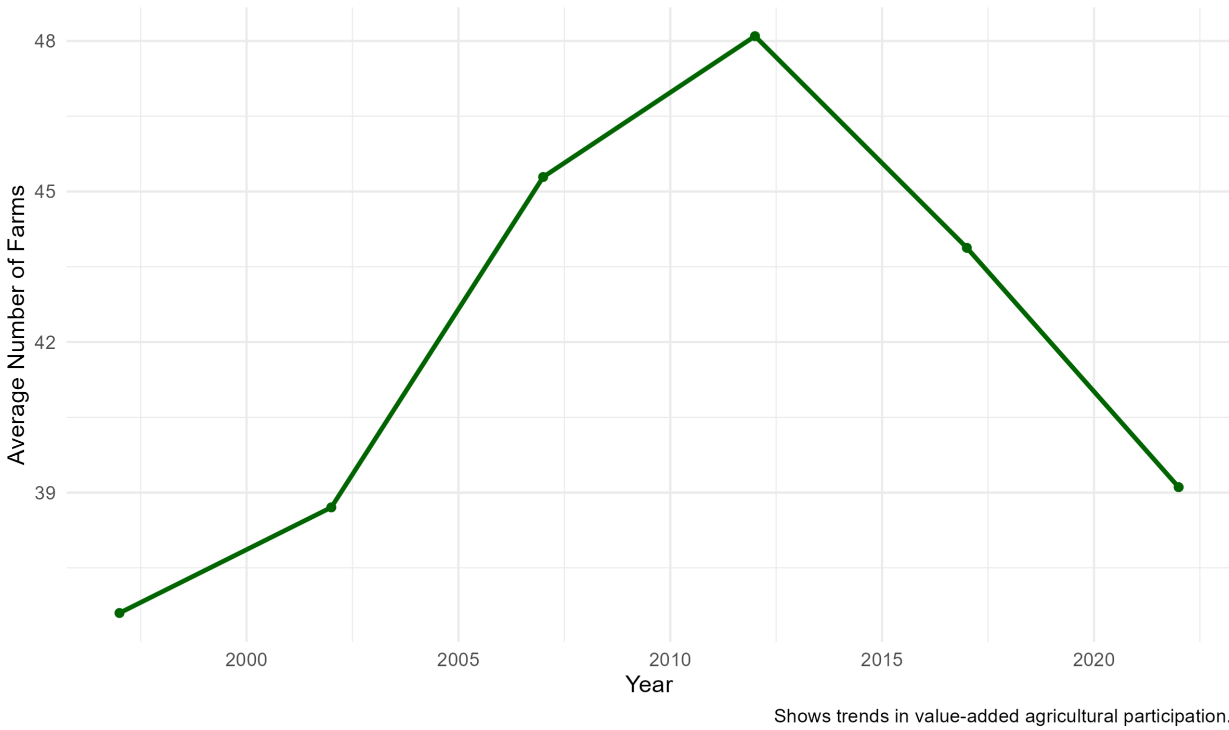


Figure 2 uses the USDA census data to measure average number of farms with value-added sales over time. A decrease can be seen from 2012 to 2017. The likely reason for this is that the variable was slightly redefined in 2017 which could have made the actually number classified on the category smaller. Interesting however the new category was very similar and say including value added so in theory it should have been a more inclusive label but not less. Regardless that likely explains the dip we see in 2017. It is interesting that we see a decrease between 2017 to 2022 however. I think the likely source of decrease here is something Covid related. Covid could have made more people less inclined to go to on farm markets as they could run the risk of exposure. Overall we see the number of farms participating in value added growing over time until that variable is redefined and then again after Covid. Again the overall outcome of interest is number of farmers markets for the farmer's market data or farms that participate in value added for the census data. This graph shows the overall trends of this outcome variable in the census data set over time.

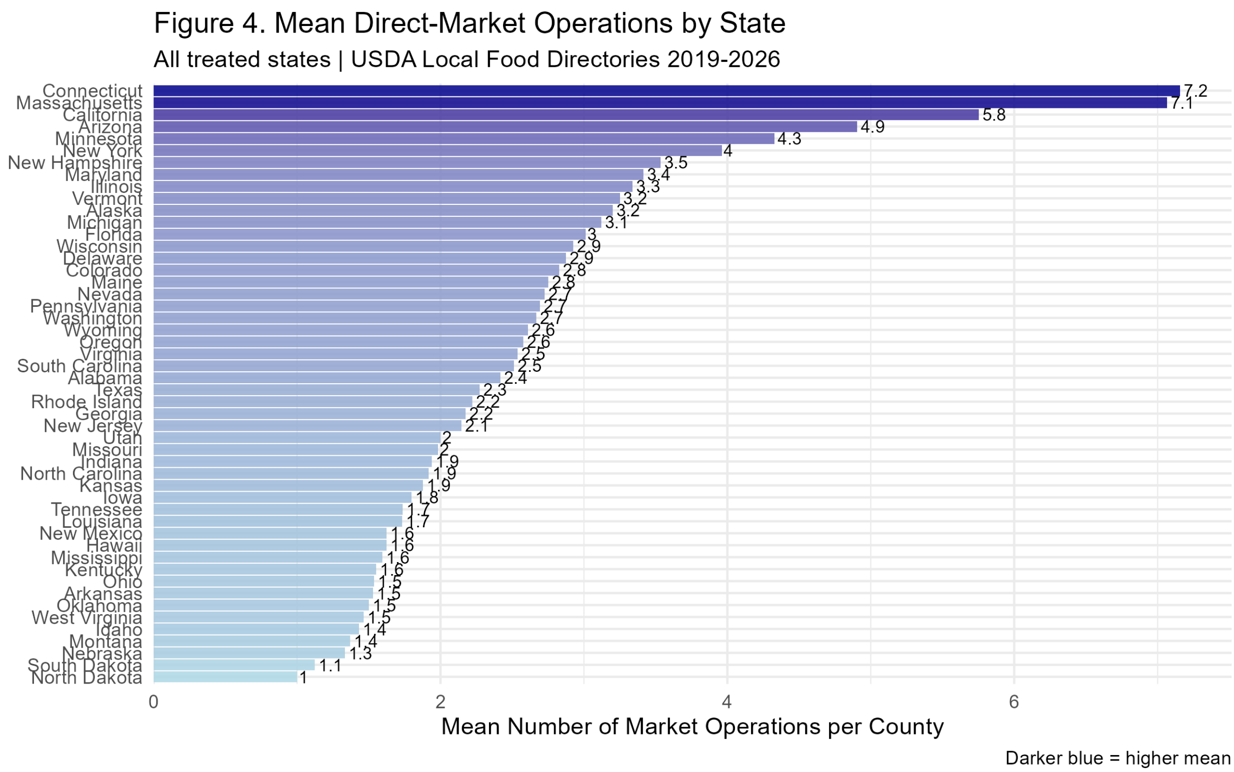
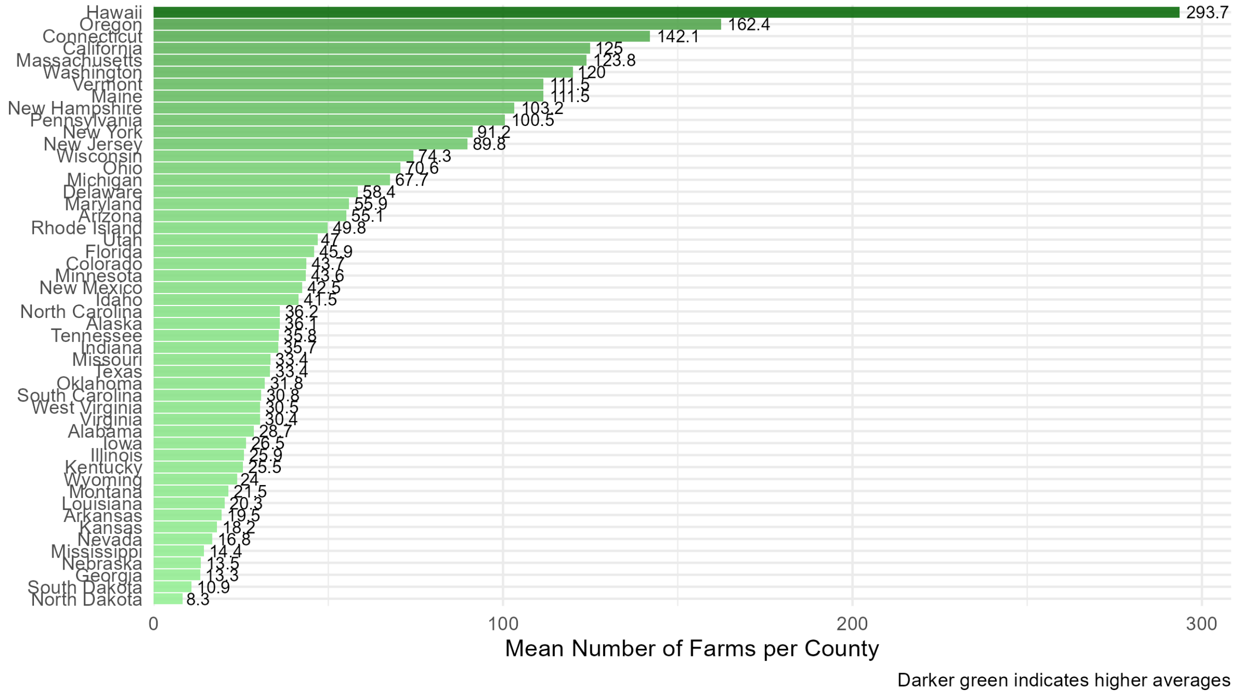


Figure 4 shows by state the mean number of operations of farmers markets. Connecticut is the state with the most during the time period provided having on average 7.2 farmers markets per county. The state with the least is North Dakota with only around 1 farmers market. This graph is able to convey information about how the outcome variable of farmer's market operations differs by state in the data. This is an important figure for determining outliers in the data set.

Figure 5. Average Number of Farms with Value-Added Sales by State

USDA Census Data (County-Level Aggregation)



This graph shows using the census data instead the average number of farms by state. Using the census data Hawaii is now one of the top adoptors Minnesota falling in the middle. This leads me to believe that many individuals in Hawaii just may not know about the program or register their market. Alternatively it could be that farmers market is not a great proxy for value added and that the Hawaii direct to consumer is not in the form of farmers market. However our states towards the bottom North Dakota and South Dakota did not change suggesting that those states truly do not participate in value added as much. Similarly the above graph is able to convey information about how the outcome variable of the number of value added farms differs by state in the data. This is an important figure for determining outliers in the data set.

Regression Results:

Table 1. Agritourism Liability Law Adoption and Local Food Market Activity (Full Sample)

	(1) Log(New Market Registrations)	(2) Log(Value-Added Farms)
Agritourism Liability Law Adopted (=1)	-0.026	0.021
	(0.215)	(0.049)
Num.Obs.	1296	18002
R2	0.670	0.905

* p < 0.1, ** p < 0.05, *** p < 0.01

Standard errors clustered at the state level in parentheses.

All models include county and year fixed effects. Outcomes are log(1 + count).

Model 1: USDA Local Food Directory, 2022-present (annual new registrations only).

Model 2: USDA Census of Agriculture, 1997-2022 (5-year intervals, total farms).

Table 1 presents the main regression results for both outcome variables using the full sample. The coefficient on the law adoption indicator in Model 1 the farmers market registration model is -0.026 with a standard error of 0.215. This estimate is not statistically different from zero at any conventional significance level suggesting that the data does not show a systematic relationship between agritourism limited liability law adoption and the rate of new farmers market registrations at the county level. The negative point estimation shouldn't be proof of an actual negative correlation given the large standard error and limitations in the data. The estimate implies that counties in law adopting states registered about 2.6% fewer new farmers markets per year than counties in non adopting states but this could be due to sampling variation only.

Model 2 using the USDA Census of Agriculture data produces a coefficient of 0.021 with a standard error of 0.049. This estimate is similarly not statistically significant at conventional levels but the point estimate is positive and the standard error is considerably smaller than in Model 1 which is reflective of the larger sample from the census data. The coefficient implies that counties in states with an agritourism limited liability law had on average about 2.1% more farms reporting value added sales per census year than counties in non adopting states after controlling for count and year fixed effects. The null can't be rejected of no association the direction of the estimate is consistent with the hypothesis that liability protection is somewhat associated with greater participation in value added agriculture. The high R-squared values in both models (0.670 and 0.905 respectively) reflect the absorptive power of the county and year fixed effects rather than the explanatory contribution of the treatment variable specifically.

Table 2. Robustness Check: Excluding High-Production Non-Adopting States (Census Model)

	(1) Full Sample	(2) Ex. Connecticut	(3) Ex. CT + CA + MA	(4) Ex. CT + CA + MA + NH
Agritourism Liability Law Adopted (=1)	0.021	0.026	-0.005	0.001
	(0.049)	(0.050)	(0.033)	(0.034)
Num.Obs.	18002	17954	17536	17476
R2	0.905	0.904	0.900	0.900

* p < 0.1, ** p < 0.05, *** p < 0.01
Dependent variable: log(1 + value-added farms per county). USDA Census of Agriculture.
Standard errors clustered at the state level in parentheses.
All models include county and year fixed effects.
Excluded states are high-production non-adopters identified in Figure 5.
Exclusions are sensitive checks; the full-sample estimate (col. 1) is the main result.

Table 2 checks if the full sample census estimate is sensitive to the inclusion of high production states that don't have agritourism liability laws. As seen in figure 5 Connecticut, California, and Massachusetts all rank among the five higher producing states in the sample by mean value added per county yet none have agritourism laws. Their presence in the control group may compress the estimated association downward by raising the baseline against which treated counties are compared.

The robustness checks reveal an interesting pattern. Excluding Connecticut alone (specification 2) moves the coefficient from 0.021 to 0.026 a slight upward shift in the expected direction. However excluding California and Massachusetts in addition to Connecticut (specification 3) causes the coefficient to turn negative at -0.005 and adding New Hampshire (specification 4) returns it to near zero at 0.001. None of these estimates are statistically significant. The instability of the coefficient across exclusion specifications conveys information. It suggests that the near zero full sample estimate is not just due to a few control states compressing a larger effect. The association between law adoption and value added farm participation appears to just be small and imprecisely estimates regardless of which states are included.

Table 3. Robustness Check: Excluding High-Production Non-Adopting States (Farmers Market Model)

	(1) Full Sample	(2) Ex. Connecticut	(3) Ex. CT + CA + MA	(4) Ex. CT + CA + MA + AZ
Agritourism Liability Law Adopted (=1)	-0.026	-0.009	0.033	0.035
	(0.215)	(0.214)	(0.210)	(0.209)
Num.Obs.	1296	1279	1161	1145
R2	0.670	0.671	0.641	0.642

* p < 0.1, ** p < 0.05, *** p < 0.01

Dependent variable: log(1 + new farmers market registrations per county). USDA Local Food Directory.

Standard errors clustered at the state level in parentheses.

All models include county and year fixed effects.

Excluded states are high-production non-adopters identified in Figure 4.

Exclusions are sensitive checks; the full-sample estimate (col. 1) is the main result.

Table 3 repeats the robustness exercise for the farmers market registration outcome. The full sample coefficient is -0.026 and excluding Connecticut alone goes it to -0.009 which is a shift towards zero. Excluding Connecticut, California, and Massachusetts together moves the coefficient to a positive value of 0.033 and adding Arizona is very similarly 0.035. This pattern is more consistent of what would be expected with controlling for high production states the coefficient would shift from negative to positive. None of the estimates are statistically significant however and the farmers market variable is subject to the registration measurement limitation so it is difficult to make strong conclusions from just the sign change. The most sensible interpretations is that there isn't a robust statistical association between law adoption and farmers market registrations rates based on this data set.

Table 4. Agritourism Liability Law Adoption and Local Food Market Activity (State-Year Panel)

	(1) Mean Log Markets	(2) Log Total Markets	(3) Mean Log Farms	(4) Log Total Farms	(5) Mean Log Markets Ex. CA/CT/MA	(6) Log Total Farms Ex. CA/CT/MA
Agritourism Liability Law Adopted (=1)	0.076	-0.035	-0.013	-0.024	0.088	-0.026
	(0.141)	(0.349)	(0.048)	(0.048)	(0.114)	(0.053)
Num.Obs.	258	258	300	300	241	282
R2	0.570	0.881	0.976	0.983	0.582	0.983

* p < 0.1, ** p < 0.05, *** p < 0.01

Unit of observation is the state-year.

All models include state and year fixed effects.

Robust standard errors in parentheses (HC1).

Models 1-2: USDA Local Food Directory, 2022-present.

Models 3-4: USDA Census of Agriculture, 1997-2022.

Models 5-6: California, Connecticut, and Massachusetts excluded.

Table 4 shows results from state year panel regressions in which the unit of observation is aggregated to the state level which matches the treatment variable assignment. This aggregation eliminates the within state county variation that is not informative for identifying the treatment effect and avoids the need to cluster standard errors at the state level. Across all six specifications the coefficients on law adoption are small and not statistically significant. Point estimates range from -0.035 to 0.088 from the farmers market outcomes and from -0.024 to -0.013 for the census outcomes in the full sample with all standard errors large relative to the estimates. The state level results are broadly consistent with the county level findings that there is no statistically detectable association between agritourism limited liability law adoption and local food market activity in either data set at any level of aggregation. The extremely high R-squared values in the census model (0.976 and 0.983) again reflect fixed effects absorption rather than treatment effect precision and shouldn't be interpreted as evidence that the model fits well in a predictive sense.

Results:

The estimated coefficient on the agritourism liability law indicator is small and statistically indistinguishable from zero across specifications. In the preferred fixed effects model, adoption of a law is associated with approximately a 2–3% decrease in new farmers market registrations, though the estimate is imprecise and not statistically significant at conventional levels.

This suggests that, within the variation observed in the data, there is no strong evidence of a systematic relationship between agritourism liability protections and the short-run formation of new farmers markets. While the negative point estimate may initially seem counterintuitive, its lack of statistical significance implies that it is not meaningfully different from zero and should not be overinterpreted.

Results using Census measures of farm activity similarly show no statistically significant relationship between agritourism liability laws and the number of value-added or direct-to-consumer farms. Across specifications, coefficient estimates are close to zero and estimated with substantial uncertainty.

Taken together with the farmers market results, this consistency suggests that the absence of detectable effects is not driven by a particular outcome measure, but rather reflects a broader lack of strong correlation between these policies and observed measures of agritourism-related activity.

Robustness checks excluding early-adopting or large states (e.g., California, Massachusetts, Connecticut) yield qualitatively similar results. The estimated coefficients remain small and statistically insignificant, suggesting that the main findings are not driven by a small number of influential states or outliers in the policy adoption process.

There are several plausible explanations for these null results. First, agritourism liability laws may primarily affect longer-run business decisions, while the outcomes considered here—particularly new market registrations—capture relatively short-run dynamics. Second, the available outcome measures may imperfectly proxy for agritourism activity. For example, farmers markets represent only one channel of direct-to-consumer sales and may not fully reflect broader agritourism participation.

Additionally, policy adoption is unlikely to be random. States that pass agritourism laws may differ systematically from those that do not, for example in terms of rural economic conditions, tourism demand, or agricultural structure. These differences may obscure any true causal relationship, particularly in a setting where the identifying variation is limited.

As a result, while the regression analysis provides useful descriptive evidence, the estimates should not be interpreted causally. Instead, they suggest that any relationship between agritourism liability protections and observable measures of farm or market activity is either small, delayed, or difficult to detect with the available data. Overall the evidence suggests that any relationship between agritourism limited liability laws and DTC agricultural activity is economically small relative to underlying geographic and temporal variation captured by fixed effects.

Causal Inference and Limitations:

The TWFE specification controls for stable county differences and common time trends but can't fix the fundamental issue that adoption of agritourism liability laws is likely endogenous to the outcomes being studied. States with growing DTC sectors may be more likely to adopt these laws which would bias beta upward. At the same time the concentration of high production states without laws in the control groups may bias beta downward. These two forces work in opposite directions making the net direction of bias uncertain and hard to get at without further model specifications.

This question also relies on a large assumption that there is truly a clustering effect of agritourism and DTC or that these markets in some way are interconnected. If this is indeed not the case then the results seen make perfect sense as limited liability agritourism laws should not have an effect on DTC if it is not associated with agritourism.

Additionally the project is limited by the USDA Local Food Directory records only stating new registrations per year not total active markets. As seen in the visualizations new registrations spiked sharply in 2023 which is likely a result of a push by the AMS rather than a genuine surge in markets opening. This means the farmers market model may capture some administrative patterns rather than true change in DTC activity. Therefore the census model results are probably a more accurate measure of actual count.

The 5 year frequency of the Census of Agriculture also limits the precision with which treatment timing can be captured. A state adopting a law between years will have all within interval variation in outcomes unobserved attenuating estimates towards zero. However the argument can be made that because implementation of DTC is a long term investment as long as we have a post period measure for each the effect should be captured as the question looks at changing of the structure of the markets long term. That being said the census model may still understate any true association even before accounting for the other sources of bias. The farmers market regression sample is also greatly reduced by the singleton problem where about 70% of counties appear in only one year of the Local Food Directory data and are therefore dropped by the fixed effects estimator leaving under 1300 observations.

Moving Forward:

If this project were to move toward a more credible causal answer the most productive next step would be to obtain annual state level data on DTC farm sales such as from the ARMS data set to replace the 5 year interval census data. With annual data an event study around law adoption dates would be feasible to look at a visual and statistical test of the parallel trends assumption and if outcomes diverge right after adoption.

Longer term the goal is to connect agritourism limited liability laws to farm survival rates among beginning and small scale operations. The motivation articulated at the outset of this project is that value added agriculture may help new farmers stay viable in the early and most financially shaky years of operation. Testing whether legal protections for agritourism translate into higher survival rates for beginning farms would require linking the law panel to individual farm level data such as from the ARMS data set.